

CYBERSNOVA



PROJECT REPORT

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Field	Details
Full Name	Namrata Ravindra Pawar
Registration Number	CSN-SP-2026-155
Internship Domain	Cyber Security
Duration	1 month
Start Date	1 Sept 2026
End Date	30 Sept 2026
Mode	30 Sept 2026
Assigned Mentor	Internship Coordinator

Level 1=Beginner
Level

PROJECT 3: LOGIN TRAFFIC ANALYSIS

2. Objective

To capture and analyze the network traffic generated during a login session on the vulnerable web application. The objective is to examine the captured packets and determine whether sensitive login information is transmitted securely, identify potential security weaknesses, and understand the security implications of transmitting authentication data over the network.

3. Target

<http://testasp.vulnweb.com/>

4. Tools Used

- Wireshark,
- Burp Suite Community Edition

Target & Test Setup

- ***Target & Test Setup:***

The login traffic analysis was performed against the authorized vulnerable web application:

- ***Target :***

<http://testasp.vulnweb.com/login.asp>

A test login session was generated using dummy credentials. Wireshark was used to capture the network traffic, while Burp Suite Community Edition was used to inspect the HTTP request and response generated during the login attempt.

The login attempt resulted in an “Invalid login” response. A successful login was not required for this traffic-analysis exercise.

Wireshark Traffic Capture

- **Wireshark Traffic Capture**

Wireshark was used to capture the network traffic generated during the login attempt. The traffic was filtered to identify HTTP POST requests associated with the login page.

The captured traffic showed a request to: `POST /login.asp HTTP/1.1`

The request was sent to the target web server over HTTP on port 80. The login form used the parameters `tfUName` and `tfUPass`.

- **Observation:**

The login request was transmitted using HTTP rather than HTTPS, indicating that TLS encryption was not being used for the login traffic.

Screenshot: Insert your Wireshark screenshot showing the POST /login.asp request here.

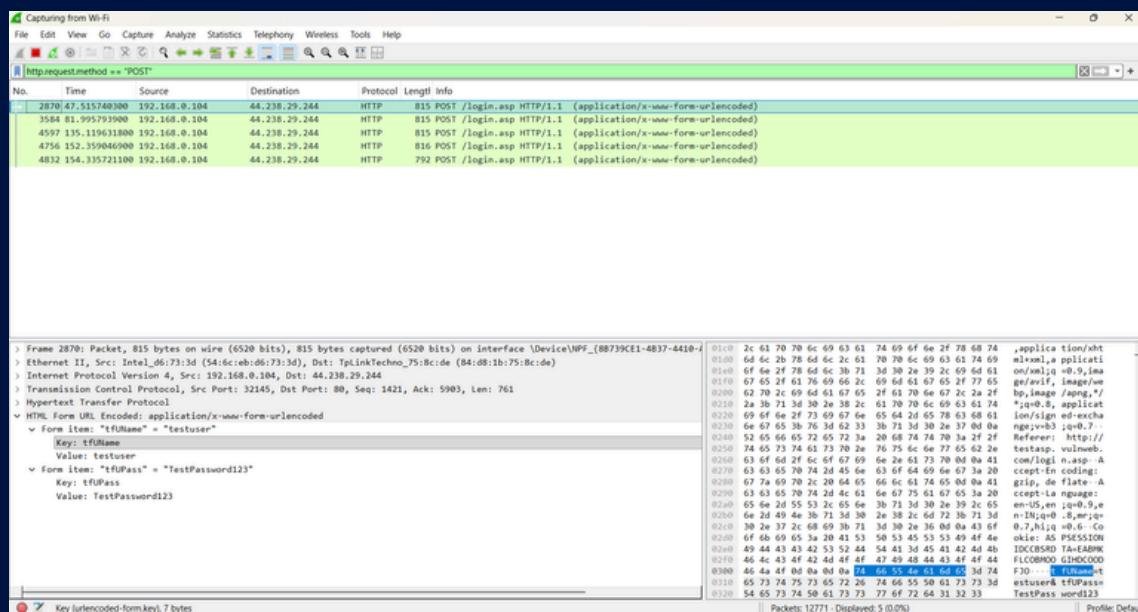


Figure 1: Wireshark capture of the login POST request

Wireshark HTTP Stream Analysis

- **HTTP Stream Analysis:**

The captured login request was further analyzed using Wireshark's Follow → HTTP Stream feature. The HTTP stream showed the communication between the client and the vulnerable web application during the login attempt.

The login form submission was sent using the HTTP POST method with the form fields tfUName and tfUPass. Because the communication was performed over HTTP rather than HTTPS, the contents of the request were not protected by TLS encryption.

The server responded with an “Invalid login” message, confirming that the test credentials were rejected.

- **Observation:**

he analysis demonstrates that authentication data can be exposed when transmitted through an unencrypted HTTP connection.

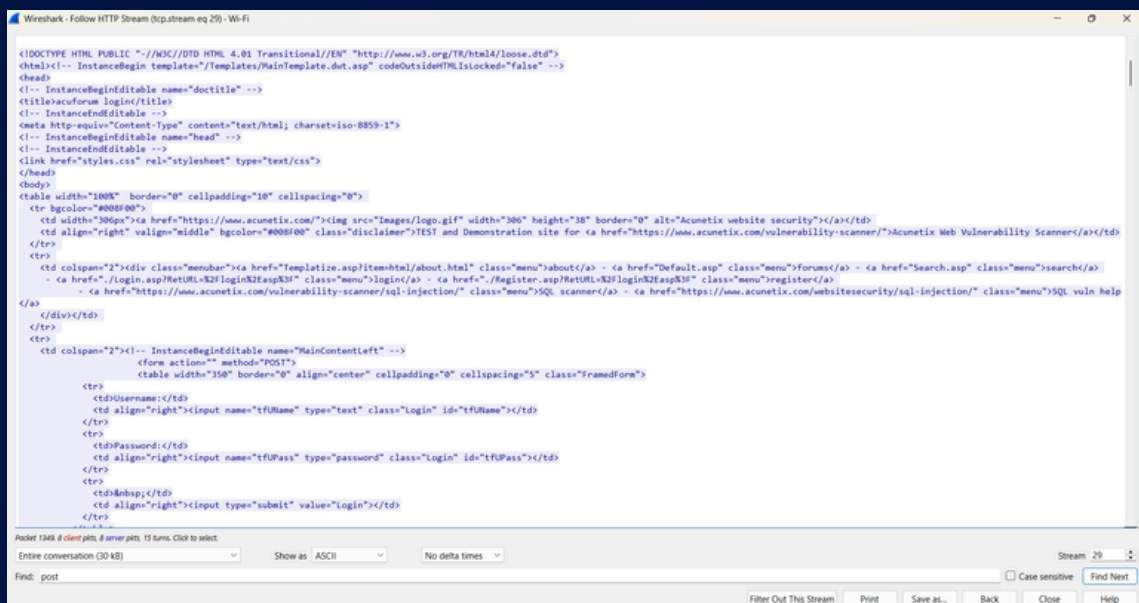


Figure 2: Wireshark HTTP Stream showing the login communication

Burp Suite Analysis

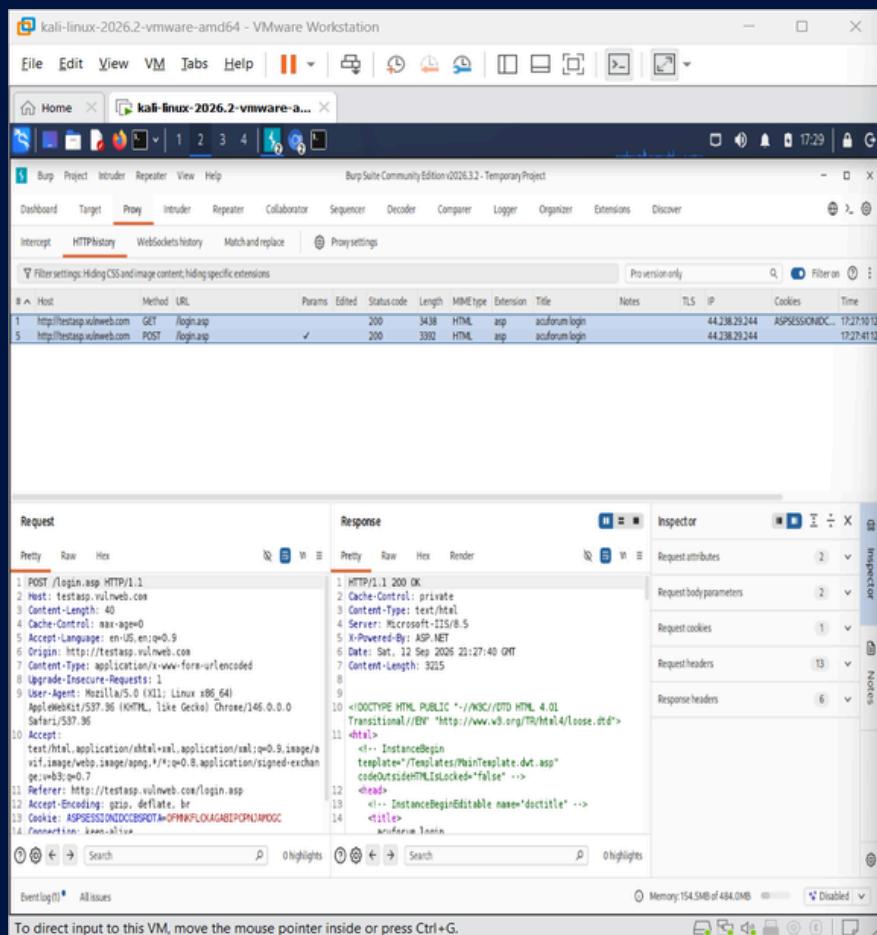
Burp Suite Community Edition was used to inspect the HTTP traffic generated during the login attempt. The request was observed in Proxy → HTTP history.

- The captured request was: POST/login.asp

The request used the content type application/x-www-form-urlencoded and contained the login form fields tfUName and tfUPass.

The TLS column did not indicate TLS protection for the request, confirming that the login communication was performed over HTTP rather than HTTPS.

The server returned an HTTP 200 response containing an invalid-login message because the test credentials were rejected.



Burp Suite inspection of the login POST request

Security Finding

- **Security Finding: Cleartext Transmission of Login Credentials:**

The analysis identified that the login form communicates with the web application over HTTP instead of HTTPS. The login request is sent using the HTTP POST method, and the submitted authentication fields are transmitted as form data without TLS encryption.

This creates a potential credential exposure risk because an attacker capable of monitoring the network traffic could potentially observe sensitive authentication information.

- **Potential Impact:**

1. Exposure of login credentials through network traffic.
2. Risk of credential theft on an untrusted or compromised network.
3. Increased possibility of account compromise if exposed credentials are reused elsewhere.

- **Recommended Mitigation:**

1. Use HTTPS/TLS for all authentication and sensitive communications.
2. Redirect HTTP requests to HTTPS.
3. Configure secure cookies and appropriate security headers.
4. Avoid transmitting authentication information over unencrypted connections.

Result

Test/Analysis	Result	Observation
Target access	Successful	Login page was accessible
Wireshark capture	Successful	Login traffic was captured
HTTP method	POST	Login data was submitted using POST
Transport protocol	HTTP	Traffic used unencrypted HTTP
Login fields	tfUName, tfUPass	Authentication fields were present in the
Burp HTTP History	Successful	POST /login.asp was captured
TLS protection	Not used	No HTTPS/TLS protection was
Login result	Invalid login	Test credentials were rejected

- **Overall Result:**

The analysis confirmed that the vulnerable application's login communication was performed over unencrypted HTTP. The use of HTTP instead of HTTPS presents a security weakness because sensitive authentication information is not protected by TLS during transmission.

Conclusion

The login traffic of the authorized vulnerable web application was successfully captured and analyzed using Wireshark and Burp Suite Community Edition. The analysis identified that the login request was transmitted over HTTP rather than HTTPS.

The captured POST request contained the login form fields, and the absence of TLS protection indicates that sensitive authentication information is not encrypted during transmission. The test login was rejected with an invalid-login response, but a successful login was not required to identify the transport-security weakness.

This exercise demonstrated the importance of using HTTPS/TLS to protect authentication traffic and prevent sensitive information from being exposed during network communication.